Assessment 3

DDD 2: Introduction to Programming and Basic Data Analysis

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within that zone. Outliers are present in all zones, suggesting that there are some households with significantly higher or lower incomes than the majority. 7
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Setup

```
rm(list = ls())
```

Load Required Libraries

```
# Library loading
library(tibble)
library(dplyr)
library(rlang)
library(lobstr)
library(rio)
library(ggplot2)
library(purrr)
library(labelled)
library(stargazer)
library(scales)
library(janitor)
library(flextable)
```

IHDS Data exercise

```
library(tidyverse)
library(rio)
HH_DATA <- import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/HH_DATA.dta")

IHDS<- subset(HH_DATA, select= c(ID18C, ID13, URBAN2011))
view(IHDS)
```

Assessment Environment

```
#All assessment answers and variables stored here in this environment
assessment_3 <- new.env()
```

Question 1: Data Cleaning and Visualization

Step 1

```
assessment_3$raw_data   <-   import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/
HH_DATA.dta")
```

Step 2

```
# STEP 2: Select required variables
library(tidyverse)
assessment_3$df <- assessment_3$raw_data |>
  select(ID11, ID13, STATEID, INCOMEPC) |>
  rename(
    region = ID11,
    caste = ID13,
    state_id = STATEID,
    income_pc = INCOMEPC)|> filter(!is.na(state_id))
```

Step 3 : Create new variable for geographical zones

```
assessment_3$df <- assessment_3$df |>
  mutate(
    geo_zone = case_when(
      state_id %in% c(2, 3, 6, 7, 12, 5, 22) ~ "North",
      state_id %in% c(8, 24) ~ "West",
      state_id %in% c(18, 23) ~ "Central",
      state_id %in% c(29, 33, 36, 28, 32) ~ "South",
      state_id %in% c(19, 10, 20, 21, 4) ~ "East",
      TRUE ~ NA_character_))
```

Step 4 : Clean data by removing rows with missing values in key variables

```
assessment_3$df_clean <- assessment_3$df |>
  filter(!is.na(geo_zone), !is.na(caste), !is.na(income_pc))
```

Step 5 : Contingency table - Caste vs Zones

```
assessment_3$df_clean |>
  mutate(across(c(caste, geo_zone), as.factor)) |>
  tabyl(geo_zone, caste) |>
```

```
  adorn_percentages("row") |>
  adorn_pct_formatting() |>
  janitor::adorn_ns() |>
  flextable()
```

geo_zone	1	2	3	4	5	6
Central	5.8% (236)	20.7% (849)	38.9% (1,593)	18.1% (741)	15.8% (646)	0.9% (35)
East	5.4% (377)	26.5% (1,848)	33.4% (2,327)	24.4% (1,699)	9.7% (676)	0.7% (50)
North	8.0% (628)	27.9% (2,181)	27.4% (2,142)	29.2% (2,285)	7.4% (577)	0.2% (17)
South	1.8% (174)	12.8% (1,232)	55.9% (5,366)	20.2% (1,940)	5.4% (521)	3.9% (370)
West	4.7% (216)	20.6% (949)	47.4% (2,181)	17.4% (800)	8.7% (398)	1.2% (54)

#Step 6 : Summary statistics of income by zone

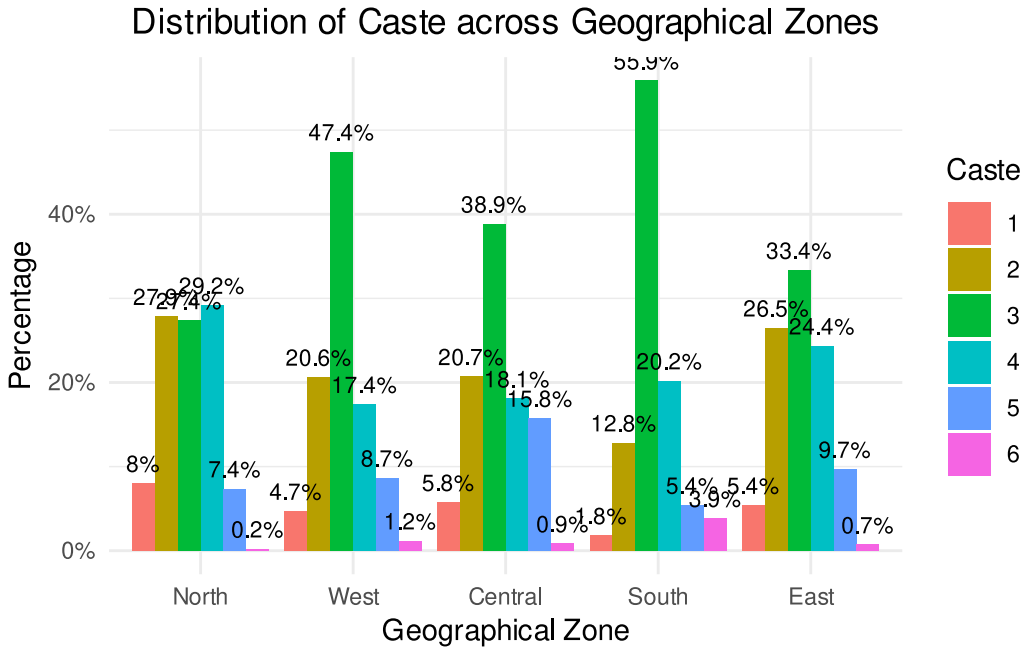
```
assessment_3$df_clean |>
  group_by(geo_zone) |>
  summarise(
    n = n(),
    mean_income = mean(income_pc, na.rm = TRUE),
    min_income = min(income_pc, na.rm = TRUE),
    max_income = max(income_pc, na.rm = TRUE))
```

```
# A tibble: 5 × 5
  geo_zone     n mean_income min_income max_income
  <chr>    <int>       <dbl>      <dbl>      <dbl>
1 Central   4100      21229.    -19800     770333.
2 East      6977      23900.    -22726.   2792455
3 North     7830      39457.    -45366.   4161000
4 South     9603      31070.   -289008.   2698850
5 West      4598      28324.   -259260    1650000
```

PLOT 1: Caste distribution across zones

```
assessment_3$df_clean |>
  filter(!is.na(caste), !is.na(geo_zone)) |>
```

```r
count(geo_zone, caste) |>
group_by(geo_zone) |>   # <-- important fix
mutate(prop = n / sum(n),top_lab = paste0(round(prop * 100, 1), "%")) |>
ungroup() |>
ggplot(aes(
 x = factor(geo_zone, levels = c("North", "West", "Central", "South", "East")),
  y = prop,
  fill = factor(caste))) +
geom_col(position = position_dodge(width = 0.9)) +
geom_text(
  aes(label = top_lab),
  position = position_dodge(width = 0.9),
  vjust = -0.5,
  size = 3) +
scale_y_continuous(labels = scales::percent) +
labs(
  x = "Geographical Zone",
  y = "Percentage",
  title = "Distribution of Caste across Geographical Zones",
  fill = "Caste") +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5),
  legend.position = "right")
```

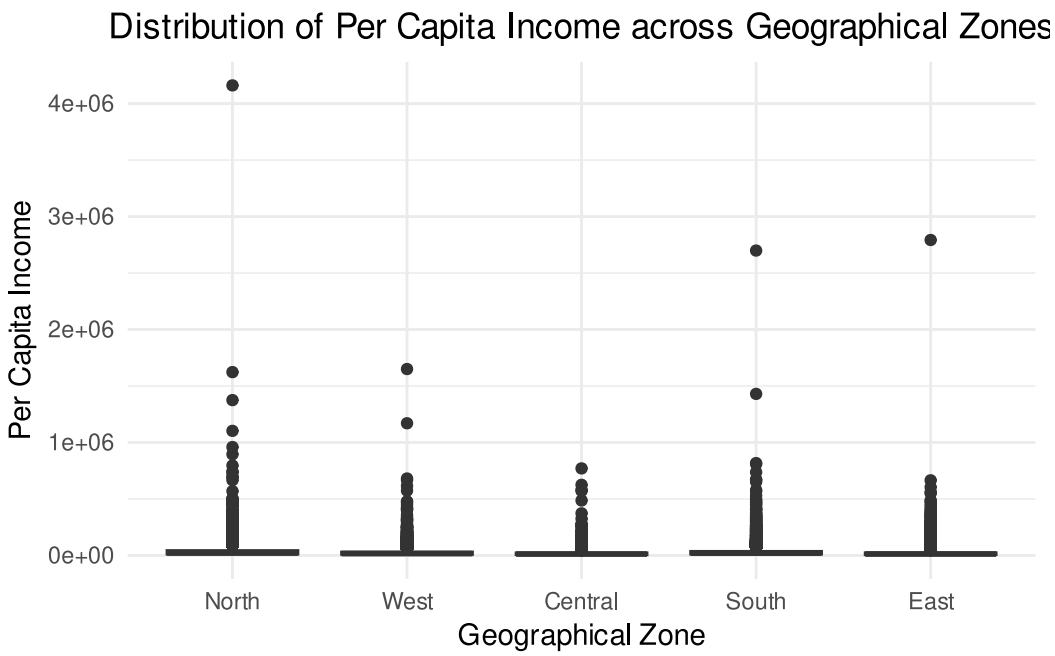


Interpretatiion of the above plot:

The bar plot illustrates the distribution of caste categories across different geographical zones. Each bar represents a zone, and the segments within the bars represent the proportion of each caste category within that zone. The percentages displayed on top of each segment indicate the proportion of that caste category in relation to the total population of the respective zone. From the plot, we can observe that certain zones have a higher concentration of specific caste categories. For instance, the North zone has a higher percentage of General caste, while the South zone has a higher percentage of OBC. The Central and East zones show a more balanced distribution across caste categories. This visualization helps us understand the demographic composition of each geographical zone in terms of caste distribution.

PLOT 2 : plot the Distribution of per capita income across zones

```
assessment_3$df_clean |>
  filter(!is.na(geo_zone), !is.na(income_pc)) |>filter(income_pc>0) |>
  ggplot(aes(x = factor(geo_zone, levels = c("North", "West", "Central", "South",
"East")), y = income_pc)) +
  geom_boxplot() +
  labs(
    x = "Geographical Zone",
    y = "Per Capita Income",
    title = "Distribution of Per Capita Income across Geographical Zones"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```



Interpretation of the above plot:

The boxplot illustrates the distribution of per capita income across different geographical zones. The median income is represented by the line within each box, while the boxes themselves show the interquartile range (IQR). The whiskers extend to 1.5 times the IQR, and any points beyond that are considered outliers. From the plot, we can observe that the North and South zones have higher median incomes compared to the Central and East zones. Additionally, the West zone shows a wider spread of income, indicating greater variability in per capita income within that zone. Outliers are present in all zones, suggesting that there are some households with significantly higher or lower incomes than the majority.

Final Check

```
ls(assessment_3)
```

```
[1] "df"       "df_clean" "raw_data"
```